

week_8_lab_tenancy

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This file cleans the quarterly tenancy bond dataset: converting rent columns to numeric, filtering to the same time window as the Airbnb dataset, and dropping rows with missing/national-total Location Id, ready for comparison with Airbnb data

lab instruction: Filter the timeframe to cover the same months as the other dataset. Clean the bond dataset, too. Similar rules as above apply - you decide what makes sense, document well, and push the final code to your repository. Please make sure to keep the Location Id and the TimeFrame. Next week we will be to combine both datasets and compare rental prices and number of available properties.

```
## # A tibble: 6 x 12
##   TimeFrame 'Location Id' 'Dwelling Type' 'Number Of Beds' 'Total Bonds'
##   <date>    <chr>        <chr>         <chr>          <dbl>
## 1 2026-04-01 NULL          ALL             1             2064
## 2 2026-04-01 -99          ALL             1            22014
## 3 2026-04-01 100100        ALL             1              6
## 4 2026-04-01 100600        ALL             1              6
## 5 2026-04-01 100700        ALL             1              6
## 6 2026-04-01 100800        ALL             1             15
## # i 7 more variables: 'Active Bonds' <dbl>, 'Closed Bonds' <dbl>,
## #   'Median Rent' <chr>, 'Geometric Mean Rent' <chr>,
## #   'Upper Quartile Rent' <chr>, 'Lower Quartile Rent' <chr>,
## #   'Log Std Dev Weekly Rent' <chr>

## [1] "TimeFrame"          "Location Id"
## [3] "Dwelling Type"      "Number Of Beds"
## [5] "Total Bonds"        "Active Bonds"
## [7] "Closed Bonds"       "Median Rent"
## [9] "Geometric Mean Rent" "Upper Quartile Rent"
## [11] "Lower Quartile Rent" "Log Std Dev Weekly Rent"

## Rows: 226,080
## Columns: 12
## $ TimeFrame          <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-
## $ 'Location Id'      <chr> "NULL", "-99", "100100", "100600", "100700", ~
## $ 'Dwelling Type'    <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds'   <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", ~
## $ 'Total Bonds'      <dbl> 2064, 22014, 6, 6, 6, 15, 6, 6, 9, 9, 12, 12~
## $ 'Active Bonds'     <dbl> 19563, 113820, 9, 12, 30, 51, 36, 21, 33, 27~
## $ 'Closed Bonds'     <dbl> 1176, 6969, 0, 3, 3, 0, 3, 3, 3, 3, 3, 0, ~
## $ 'Median Rent'      <chr> "NULL", "550", "425", "588", "550", "500", "~
## $ 'Geometric Mean Rent' <chr> "NULL", "522", "408", "585", "530", "520", "~
## $ 'Upper Quartile Rent' <chr> "NULL", "660", "475", "720", "579", "550", "~
```

```
## $ 'Lower Quartile Rent'      <chr> "NULL", "450", "370", "545", "473", "485", "~
## $ 'Log Std Dev Weekly Rent' <chr> "NULL", "0.4144", "0.2834", "0.255", "0.1588~
```

```
##      TimeFrame      Location Id      Dwelling Type      Number Of Beds
## Min.   :2020-01-01   Length   :226080   Length   :226080   Length   :226080
## 1st Qu.:2021-07-01   N.unique : 2011   N.unique :    6   N.unique :   13
## Median :2023-04-01   N.blank  :    0   N.blank  :    0   N.blank  :    0
## Mean   :2023-03-07   Min.nchar:    3   Min.nchar:    3   Min.nchar:    1
## 3rd Qu.:2025-01-01   Max.nchar:    6   Max.nchar:   14   Max.nchar:    3
## Max.   :2026-04-01                                     NAs      : 12009
##      Total Bonds      Active Bonds      Closed Bonds      Median Rent
## Min.    : 6.00   Min.    : 0.0   Min.    : 0.00   Length   :226080
## 1st Qu.: 6.00   1st Qu.: 48.0   1st Qu.: 3.00   N.unique : 1029
## Median : 9.00   Median : 93.0   Median : 9.00   N.blank  :    0
## Mean    : 33.81  Mean    : 358.4   Mean    : 27.97  Min.nchar:    1
## 3rd Qu.: 15.00  3rd Qu.: 165.0   3rd Qu.: 15.00  Max.nchar:    4
## Max.    :49875.00 Max.    :520821.0 Max.    :48384.00
## Geometric Mean Rent Upper Quartile Rent Lower Quartile Rent
## Length   :226080   Length   :226080   Length   :226080
## N.unique : 1435   N.unique : 1459   N.unique : 1141
## N.blank  :    0   N.blank  :    0   N.blank  :    0
## Min.nchar:    1   Min.nchar:    1   Min.nchar:    1
## Max.nchar:    4   Max.nchar:    4   Max.nchar:    4
##
## Log Std Dev Weekly Rent
## Length   :226080
## N.unique : 8923
## N.blank  :    0
## Min.nchar:    1
## Max.nchar:    6
##
```

check how many literal “NULL” text strings exist versus actual missing values (NA/NaN) across the dataset

```
##      TimeFrame      Location Id      Dwelling Type
##      0              794              0
##      Number Of Beds      Total Bonds      Active Bonds
##      0              0              0
##      Closed Bonds      Median Rent      Geometric Mean Rent
##      0              803              803
##      Upper Quartile Rent      Lower Quartile Rent      Log Std Dev Weekly Rent
##      803              803              859
```

TimeFrame 0 Location Id 794 <- has Null strings ?should be numeric Dwelling Type 0 Number Of Beds 0 Total Bonds 0 Active Bonds 0 Closed Bonds 0 Median Rent 803 <- has Null strings Geometric Mean Rent 803 <- has Null strings Upper Quartile Rent 803 <- has Null strings Lower Quartile Rent 803 <- has Null strings Log Std Dev Weekly Rent 859 <- has Null strings

Check location id: does it contain any other non-numeric text besides “NULL” and “-99”

```
## character(0)
```

Location Id contains no other non-numeric text besides NA/NULL and -99 NULL: The government deliberately hid or left out that data because there weren’t enough rentals there to protect people’s privacy. -99:

Code the government uses to show “the total for the whole country of New Zealand combined,” not a specific city.

Need to decide what to do with “NULL” strings and -99 in location ID columns?

Check other columns:

```
## [1] "ALL"           "Apartment"      "Boarding House" "Flat"
## [5] "House"         "Room"

## [1] "1"  "2"  "3"  "4"  "5"  "9"  "ALL" "5+" "6"  NA  "0"  "7"
## [13] "15" "8"

## Rows: 226,080
## Columns: 12
## $ TimeFrame          <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-~
## $ 'Location Id'     <chr> "NULL", "-99", "100100", "100600", "100700",~
## $ 'Dwelling Type'   <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds'  <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",~
## $ 'Total Bonds'     <dbl> 2064, 22014, 6, 6, 6, 15, 6, 6, 9, 9, 12, 12~
## $ 'Active Bonds'    <dbl> 19563, 113820, 9, 12, 30, 51, 36, 21, 33, 27~
## $ 'Closed Bonds'    <dbl> 1176, 6969, 0, 3, 3, 0, 3, 3, 3, 3, 3, 0,~
## $ 'Median Rent'     <chr> "NULL", "550", "425", "588", "550", "500", "~
## $ 'Geometric Mean Rent' <chr> "NULL", "522", "408", "585", "530", "520", "~
## $ 'Upper Quartile Rent' <chr> "NULL", "660", "475", "720", "579", "550", "~
## $ 'Lower Quartile Rent' <chr> "NULL", "450", "370", "545", "473", "485", "~
## $ 'Log Std Dev Weekly Rent' <chr> "NULL", "0.4144", "0.2834", "0.255", "0.1588~
```

- [1] “TimeFrame” <- correct format
- [2] “Location Id” <- think later how to deal
- [3] “Dwelling Type” <- correct format
- [4] “Number Of Beds” <- categorical column because has “ALL” and “5+” along with numbers. doesn’t need fixing.
- [5] “Total Bonds” <- ok
- [6] “Active Bonds” <- ok
- [7] “Closed Bonds” <- ok
- [8] “Median Rent”<- need turn “Null” strings to NA, and covert to numeric datatype
- [9] “Geometric Mean Rent”<- need turn “Null” strings to NA, and covert to numeric datatype
- [10] “Upper Quartile Rent”<- need turn “Null” strings to NA, and covert to numeric datatype
- [11] “Lower Quartile Rent”<- need turn “Null” strings to NA, and covert to numeric datatype
- [12] “Log Std Dev Weekly Rent”<- need turn “Null” strings to NA, and covert to numeric datatype

convert all the rent columns (8-12) into proper numeric values (while automatically turning those “NULL” text strings into clean NA missing values) and keeping Location Id untouched

```
## Rows: 226,080
## Columns: 12
## $ TimeFrame          <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-~
## $ 'Location Id'     <chr> "NULL", "-99", "100100", "100600", "100700",~
## $ 'Dwelling Type'   <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds'  <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",~
## $ 'Total Bonds'     <dbl> 2064, 22014, 6, 6, 6, 15, 6, 6, 9, 9, 12, 12~
## $ 'Active Bonds'    <dbl> 19563, 113820, 9, 12, 30, 51, 36, 21, 33, 27~
## $ 'Closed Bonds'    <dbl> 1176, 6969, 0, 3, 3, 0, 3, 3, 3, 3, 3, 0,~
```

```
## $ 'Median Rent' <dbl> NA, 550, 425, 588, 550, 500, 570, 698, 580, ~
## $ 'Geometric Mean Rent' <dbl> NA, 522, 408, 585, 530, 520, 536, 649, 482, ~
## $ 'Upper Quartile Rent' <dbl> NA, 660, 475, 720, 579, 550, 621, 739, 590, ~
## $ 'Lower Quartile Rent' <dbl> NA, 450, 370, 545, 473, 485, 490, 575, 390, ~
## $ 'Log Std Dev Weekly Rent' <dbl> NA, 0.4144, 0.2834, 0.2550, 0.1588, 0.0970, ~
```

```
## TimeFrame Location Id Dwelling Type Number Of Beds
## Min. :2020-01-01 Length :226080 Length :226080 Length :226080
## 1st Qu.:2021-07-01 N.unique : 2011 N.unique : 6 N.unique : 13
## Median :2023-04-01 N.blank : 0 N.blank : 0 N.blank : 0
## Mean :2023-03-07 Min.nchar: 3 Min.nchar: 3 Min.nchar: 1
## 3rd Qu.:2025-01-01 Max.nchar: 6 Max.nchar: 14 Max.nchar: 3
## Max. :2026-04-01 NAs : 12009
```

```
## Total Bonds Active Bonds Closed Bonds Median Rent
## Min. : 6.00 Min. : 0.0 Min. : 0.00 Min. : 0
## 1st Qu.: 6.00 1st Qu.: 48.0 1st Qu.: 3.00 1st Qu.: 475
## Median : 9.00 Median : 93.0 Median : 9.00 Median : 570
## Mean : 33.81 Mean : 358.4 Mean : 27.97 Mean : 574
## 3rd Qu.: 15.00 3rd Qu.: 165.0 3rd Qu.: 15.00 3rd Qu.: 660
## Max. :49875.00 Max. :520821.0 Max. :48384.00 Max. :3350
## NAs :803
```

```
## Geometric Mean Rent Upper Quartile Rent Lower Quartile Rent
## Min. : 0.0 Min. : 0.0 Min. : 0
## 1st Qu.: 461.0 1st Qu.: 520.0 1st Qu.: 420
## Median : 558.0 Median : 620.0 Median : 518
## Mean : 564.4 Mean : 635.2 Mean : 517
## 3rd Qu.: 655.0 3rd Qu.: 728.0 3rd Qu.: 603
## Max. :2818.0 Max. :3625.0 Max. :2785
## NAs :803 NAs :803 NAs :803
```

```
## Log Std Dev Weekly Rent
## Min. :0.0000
## 1st Qu.:0.1072
## Median :0.1703
## Mean :0.2154
## 3rd Qu.:0.2765
## Max. :5.8860
## NAs :859
```

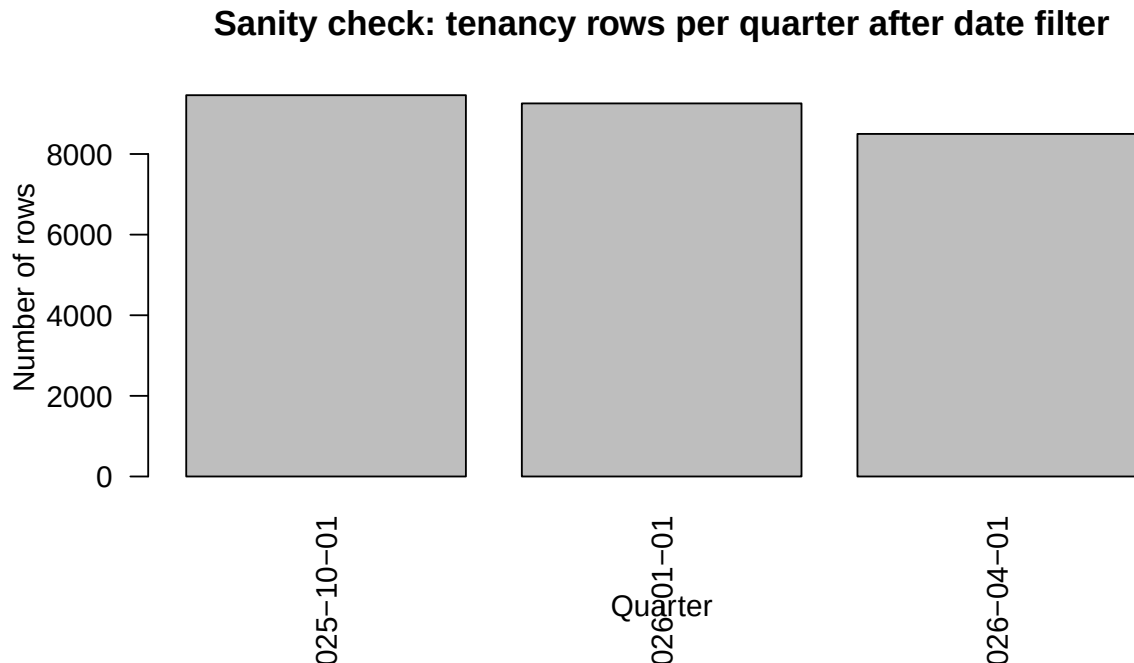
Airbnb dataset covers October 2025 to June 2026. Filter tenancy dataset so it uses the exact same timeframe window

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## "2025-10-01" "2025-10-01" "2026-01-01" "2025-12-28" "2026-04-01" "2026-04-01"
```

```
## Rows: 27,212
## Columns: 12
## $ TimeFrame <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-
## $ 'Location Id' <chr> "NULL", "-99", "100100", "100600", "100700",~
## $ 'Dwelling Type' <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds' <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1",~
## $ 'Total Bonds' <dbl> 2064, 22014, 6, 6, 6, 15, 6, 6, 9, 9, 12, 12~
## $ 'Active Bonds' <dbl> 19563, 113820, 9, 12, 30, 51, 36, 21, 33, 27~
```

```
## $ 'Closed Bonds' <dbl> 1176, 6969, 0, 3, 3, 0, 3, 3, 3, 3, 3, 3, 0, ~
## $ 'Median Rent' <dbl> NA, 550, 425, 588, 550, 500, 570, 698, 580, ~
## $ 'Geometric Mean Rent' <dbl> NA, 522, 408, 585, 530, 520, 536, 649, 482, ~
## $ 'Upper Quartile Rent' <dbl> NA, 660, 475, 720, 579, 550, 621, 739, 590, ~
## $ 'Lower Quartile Rent' <dbl> NA, 450, 370, 545, 473, 485, 490, 575, 390, ~
## $ 'Log Std Dev Weekly Rent' <dbl> NA, 0.4144, 0.2834, 0.2550, 0.1588, 0.0970, ~
```

Sanity check - confirms no quarter is empty or oversized after the date filter, by checking each quarter has a roughly similar number of rows



Summary so far: missing data stored as literal string “NULL” rather than NA for: location_id (794 “NULL” values), Median, Geometric Mean, Upper Quartile, Lower Quartile (all has 803 “NULL” values) Log Std Dev Weekly Rent: (859 “NULL” values) Convert all rent metrics (Median, Geometric Mean, Upper Quartile, Lower Quartile, Log Std Dev Weekly Rent) to numeric (automatically turns “NULL” into NA)

location_id also has -99 that represents something

left “number of beds” as categorical column because it includes strings “ALL” and “5+” alongside numbers, which may be needed for analysis

Filtered the dataset to October 2025-june 2026 as per airbnb file

```
##           TimeFrame           Location Id           Dwelling Type
##           0                0                0
##      Number Of Beds      Total Bonds      Active Bonds
##           890                0                0
##      Closed Bonds      Median Rent      Geometric Mean Rent
##           0                94                94
##      Upper Quartile Rent      Lower Quartile Rent      Log Std Dev Weekly Rent
##           94                94                94
```

```
## # A tibble: 6 x 12
```

```
##   TimeFrame 'Location Id' 'Dwelling Type' 'Number Of Beds' 'Total Bonds'
##   <date>    <chr>         <chr>         <chr>         <dbl>
## 1 2026-04-01 NULL          ALL          1             2064
## 2 2026-04-01 -99          ALL          1             22014
## 3 2026-04-01 100100       ALL          1             6
## 4 2026-04-01 100600       ALL          1             6
## 5 2026-04-01 100700       ALL          1             6
## 6 2026-04-01 100800       ALL          1             15
## # i 7 more variables: 'Active Bonds' <dbl>, 'Closed Bonds' <dbl>,
## #   'Median Rent' <dbl>, 'Geometric Mean Rent' <dbl>,
## #   'Upper Quartile Rent' <dbl>, 'Lower Quartile Rent' <dbl>,
## #   'Log Std Dev Weekly Rent' <dbl>
```

Deal with location_id column, since it's the identifier

```
## [1] 94
```

turn the "NULL" cells in location_id column into NAs

```
## [1] 94
```

Check whether the 94 missing Location Id rows are concentrated in one quarter or spread across all three.

```
## # A tibble: 3 x 2
##   TimeFrame missing_rows
##   <date>      <int>
## 1 2025-10-01      32
## 2 2026-01-01      33
## 3 2026-04-01      29
```

Each quarter has a similar number of missing rows (29-33), suggesting this is spread out by chance rather than tied to a specific time period.

Checking missing rate

```
## # A tibble: 3 x 4
##   TimeFrame total_rows missing_rows pct_missing
##   <date>      <int>      <int>      <dbl>
## 1 2025-10-01    9459         32      0.34
## 2 2026-01-01    9255         33      0.36
## 3 2026-04-01    8498         29      0.34
```

The missing rate is nearly identical across all three quarters (0.34%, 0.36%, 0.34%), confirming the missingness isn't linked to a specific time period.

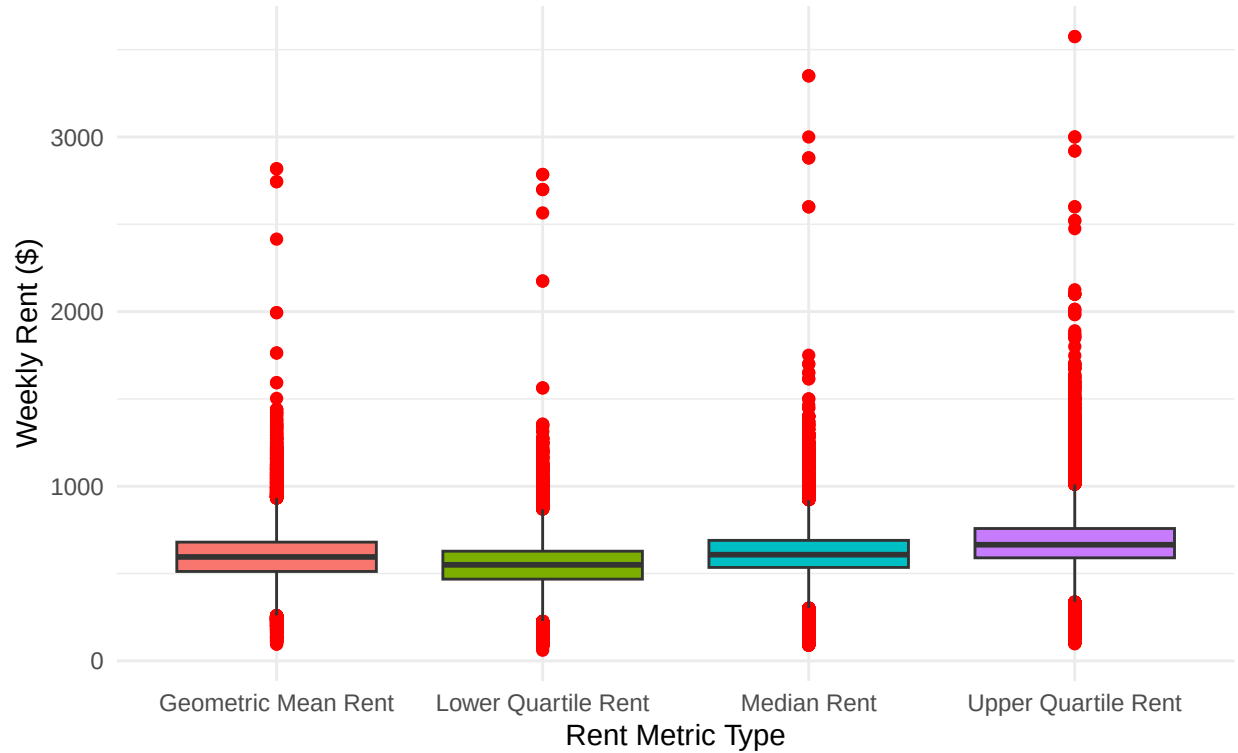
The identifier of the dataset (location id) has 94 missing values.. drop? as looks like the "NULL" were there because there were fewer than 5 bonds for that specific combination, so it's hidden so people can't trace who these tenants are. And there's no way knowing these ids.. Decision to drop these 94 rows

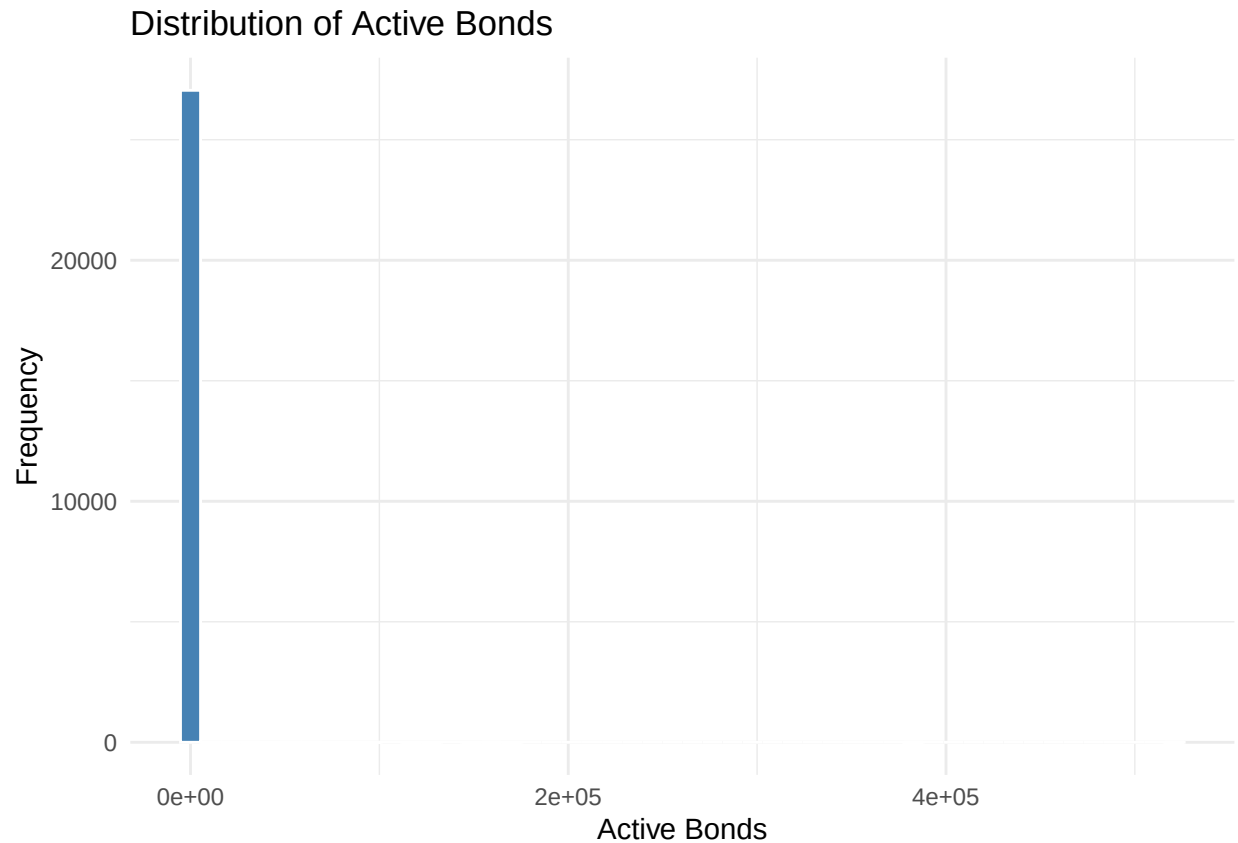
```
## [1] 0
```

##	TimeFrame	Location Id	Dwelling Type	Number Of Beds
##	Min. :2025-10-01	Length :27118	Length :27118	Length :27118
##	1st Qu.:2025-10-01	N.unique : 1819	N.unique : 6	N.unique : 11
##	Median :2026-01-01	N.blank : 0	N.blank : 0	N.blank : 0
##	Mean :2025-12-28	Min.nchar: 3	Min.nchar: 3	Min.nchar: 1
##	3rd Qu.:2026-04-01	Max.nchar: 6	Max.nchar: 14	Max.nchar: 3
##	Max. :2026-04-01			NAs : 876
##	Total Bonds	Active Bonds	Closed Bonds	Median Rent
##	Min. : 6.00	Min. : 3.0	Min. : 0.00	Min. : 90.0
##	1st Qu.: 6.00	1st Qu.: 39.0	1st Qu.: 3.00	1st Qu.: 535.0
##	Median : 9.00	Median : 87.0	Median : 6.00	Median : 608.0
##	Mean : 34.59	Mean : 358.6	Mean : 25.89	Mean : 617.3
##	3rd Qu.: 18.00	3rd Qu.: 174.0	3rd Qu.: 12.00	3rd Qu.: 690.0
##	Max. :49875.00	Max. :520821.0	Max. :48384.00	Max. :3350.0
##	Geometric Mean Rent	Upper Quartile Rent	Lower Quartile Rent	
##	Min. : 96.0	Min. : 98	Min. : 61.0	
##	1st Qu.: 512.0	1st Qu.: 590	1st Qu.: 468.0	
##	Median : 595.0	Median : 665	Median : 550.0	
##	Mean : 602.8	Mean : 689	Mean : 548.3	
##	3rd Qu.: 680.0	3rd Qu.: 758	3rd Qu.: 628.0	
##	Max. :2818.0	Max. :3575	Max. :2785.0	
##	Log Std Dev Weekly Rent			
##	Min. :0.0000			
##	1st Qu.:0.1162			
##	Median :0.1900			
##	Mean :0.2375			
##	3rd Qu.:0.3099			
##	Max. :3.1541			

Distribution of Rent Metrics across New Zealand

Red points indicate extreme outliers





```
## [1] "1" "2" "3" "4" "5" "9" "ALL" "5+" "6" NA "0" "7"

##           TimeFrame           Location Id           Dwelling Type
##                0                0                0
##      Number Of Beds      Total Bonds      Active Bonds
##            876                0                0
##      Closed Bonds      Median Rent      Geometric Mean Rent
##                0                0                0
##      Upper Quartile Rent      Lower Quartile Rent Log Std Dev Weekly Rent
##                0                0                0
```

Active bond has a mean 358.6 vs median 87, Total bond has a mean of 34.59 and median of 9. Because location id -99 are national totals so is causing this...will need to decide about this -99 Median rent has a max of 3350/week - very high min median rent of 90/week No negative or 0 rent The Number Of Beds column contains a mix of numeric values (0 to 9), text strings like 'ALL' and '5+', and missing values (NA). But currently they are all being treated as a string. -> inconsistent formatting, will need thinking about.. It also has 876 missing values remaining

Previous missing values (94) in median rent, geometric mean rent, upper quartile rent, lower quartile rent, log std dev weekly rent disappeared when dropped the location ID "NULL" rows

Rent columns see if extreme prices make sense

```
## # A tibble: 6 x 12
##   TimeFrame 'Location Id' 'Dwelling Type' 'Number Of Beds' 'Total Bonds'
##   <date>    <chr>         <chr>         <chr>         <dbl>
```

```
## 1 2026-04-01 252300      House      3      6
## 2 2026-01-01 126800      ALL        5+     6
## 3 2026-01-01 253300      ALL        5+     6
## 4 2026-01-01 143200      ALL        4      6
## 5 2026-01-01 126800      House      5+     6
## 6 2026-01-01 253300      House      5+     6
## # i 7 more variables: 'Active Bonds' <dbl>, 'Closed Bonds' <dbl>,
## #   'Median Rent' <dbl>, 'Geometric Mean Rent' <dbl>,
## #   'Upper Quartile Rent' <dbl>, 'Lower Quartile Rent' <dbl>,
## #   'Log Std Dev Weekly Rent' <dbl>
```



The extreme rents (2880-3575/week) looks plausible: All dwelling, 5+ beds, low counts in active bonds column since there's only few such properties

check duplicates:

```
## [1] 0

## # A tibble: 6 x 2
##   'Location Id'    n
##   <chr>          <int>
## 1 -99            127
## 2 100100         6
## 3 100200         4
## 4 100400         8
## 5 100600         5
## 6 100700        14
```

```
## # A tibble: 0 x 12
## # Groups:   TimeFrame, Location Id, Dwelling Type, Number Of Beds [0]
## # i 12 variables: TimeFrame <date>, Location Id <chr>, Dwelling Type <chr>,
## #   Number Of Beds <chr>, Total Bonds <dbl>, Active Bonds <dbl>,
## #   Closed Bonds <dbl>, Median Rent <dbl>, Geometric Mean Rent <dbl>,
## #   Upper Quartile Rent <dbl>, Lower Quartile Rent <dbl>,
## #   Log Std Dev Weekly Rent <dbl>

## Rows: 27,118
## Columns: 12
## $ TimeFrame                <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-
## $ 'Location Id'            <chr> "-99", "100100", "100600", "100700", "100800~
## $ 'Dwelling Type'          <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds'         <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", ~
## $ 'Total Bonds'            <dbl> 22014, 6, 6, 6, 15, 6, 6, 9, 9, 12, 12, 9, 1~
## $ 'Active Bonds'           <dbl> 113820, 9, 12, 30, 51, 36, 21, 33, 27, 69, 4~
## $ 'Closed Bonds'           <dbl> 6969, 0, 3, 3, 0, 3, 3, 3, 3, 3, 3, 0, 0, 3, ~
## $ 'Median Rent'            <dbl> 550, 425, 588, 550, 500, 570, 698, 580, 650, ~
## $ 'Geometric Mean Rent'     <dbl> 522, 408, 585, 530, 520, 536, 649, 482, 606, ~
## $ 'Upper Quartile Rent'     <dbl> 660, 475, 720, 579, 550, 621, 739, 590, 710, ~
## $ 'Lower Quartile Rent'     <dbl> 450, 370, 545, 473, 485, 490, 575, 390, 530, ~
## $ 'Log Std Dev Weekly Rent' <dbl> 0.4144, 0.2834, 0.2550, 0.1588, 0.0970, 0.19~
```

Summary: 1. Filtered the dataset to October 1, 2025 – June 30, 2026 to match the Airbnb dataset.

2. Location ID: There are 94 “NULL” rows. “NULL” is used when there are fewer than five bonds for a given selection, in order to suppress results and protect privacy. These rows had corresponding “NULL” string in five rent-related columns (Median Rent, Geometric Mean Rent, Upper Quartile Rent, Lower Quartile Rent, and Log Std Dev Weekly Rent). These 94 rows were dropped because the corresponding location IDs cannot be recovered. This has turned the rent-related column from character strings to numeric (dbl). Removing these rows also removed all NA values in the rent-related columns (Median Rent, Geometric Mean Rent, Upper Quartile Rent, Lower Quartile Rent, and Log Std Dev Weekly Rent). Question: location ID still has -99, which appears to represent country-wide total. Keeping this data makes rent right skewed (Active Bonds: mean = 358.6 vs. median = 87 Total Bonds: mean = 34.59 vs. median = 9), what to do with this?

3. Number Of Beds: contains numeric values (0–9) as well as string “ALL” and “5+” so the column is character/categorical column. Left as it is for now.
4. General findings: -No duplicates were found, either for the exact combination of variables or at the primary key level (TimeFrame, Location ID, Dwelling Type, Number of Beds). -Rent values: No negative or zero rents were found. -Rent extremes: Median rent ranged from \$90 to \$3,350 per week. The highest rent values (\$2,880–\$3,575 per week) appear plausible rather than being data-entry errors, as they are associated with 5+ bedroom/all-dwelling properties and low bond counts. Therefore, no columns or observations were dropped based on these extreme values.

Before cleaning- Rows: 226,080 Columns: 12

After cleaning- Rows: 27,118 Columns: 12

(Note Cleaned dataset: data_4)

Remove -99 rows

```
## Rows: 26,991
```

```

## Columns: 12
## $ TimeFrame          <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04-~
## $ 'Location Id'      <chr> "100100", "100600", "100700", "100800", "101~
## $ 'Dwelling Type'    <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ 'Number Of Beds'   <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "~
## $ 'Total Bonds'      <dbl> 6, 6, 6, 15, 6, 6, 9, 9, 12, 12, 9, 12, 6, 1~
## $ 'Active Bonds'     <dbl> 9, 12, 30, 51, 36, 21, 33, 27, 69, 42, 27, 2~
## $ 'Closed Bonds'     <dbl> 0, 3, 3, 0, 3, 3, 3, 3, 3, 3, 0, 0, 3, 0, 3,~
## $ 'Median Rent'      <dbl> 425, 588, 550, 500, 570, 698, 580, 650, 470,~
## $ 'Geometric Mean Rent' <dbl> 408, 585, 530, 520, 536, 649, 482, 606, 451,~
## $ 'Upper Quartile Rent' <dbl> 475, 720, 579, 550, 621, 739, 590, 710, 550,~
## $ 'Lower Quartile Rent' <dbl> 370, 545, 473, 485, 490, 575, 390, 530, 388,~
## $ 'Log Std Dev Weekly Rent' <dbl> 0.2834, 0.2550, 0.1588, 0.0970, 0.1942, 0.29~

```